Forecasting first diagnoses from synthetic EHR timelines

Research intern coding test: technical report

[AUTHOR NAME]

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Links. Code and documentation: https://github.com/Reaper8202/Research_Intern_Coding_Test. Trained model package: https://huggingface.co/Reaper8202/ehr-forecast. wandb project (training, tuning, and val-curve runs): https://wandb.ai/derek-wu820-university-of-toronto/ehr-forecast.

Summary

The submitted model is an ensemble of fifteen small decoder-only transformers (2 layers, width 128, about 738k parameters each) trained from random weights on the 2,791 training patients: three random seeds, five cross-validation folds each, self-pretrained on pre-anchor event sequences and then fully fine-tuned on the 40 first-diagnosis labels. On the official validation split (365 patients) it scores macro AUROC **0.751** [0.730, 0.773] and mAP **0.209** [0.192, 0.211], scored once by a frozen bundle. It beats the age-sex incidence reference by 0.025 AUROC (interval above zero), ranks patients about as well as logistic regression and LightGBM on code counts, and trails LightGBM on average precision by 0.03. Test scores are the organisers'.

Most of the effort went into the pipeline rather than the model: a truncation rule checked against the released test files, column allowlists that keep future-bearing fields out, a prior- condition mask shared by every model, five-fold cross-validation on train for every choice, two tuning rounds whose budgets were committed before their scores, and one recorded val run. The transformer did not beat a tuned gradient-boosted baseline on this data, and the report says so.

1. Representation

One unit of input is one clinical event token, `source:kind:code`, with its value fused in. A patient's sequence is

`[SEX:F] hx:<older conditions> ... [SEP] <events, oldest first, one token each>`

Source	Token	Example
conditions, medications, care plans, allergies, devices	`src:start:code`, `src:stop:code`	`med:start:310798`
procedures, immunizations	`proc:code`, `imm:code`	`proc:430193006`
numeric observations	`obs:code:unit:q<bin>`	`obs:8480-6:mm[Hg]:q7`
text observations	`obs:code:<normalized value>`	`obs:72166-2:never_smoker`
imaging studies, encounters	`img:modality|site`, `enc:class:code`	`enc:wellness:162673000`

Source	Token	Example
genotype (D34)	`[DNA]`, `dna:<site>:<alleles>`	one token per carried variant
brain MRI (D34)	`[IMG]`	embedding replaced by a 3-D CNN encoding

Fields and modalities. Every CSV table is read through a column allowlist. Dropped on purpose: death date, lifetime healthcare expenses and coverage, marital status, addresses, all cost and payer columns, medication dispense counts and total cost (in the released test files 873 of 1,049 medications that continue past the anchor have more dispenses than months elapsed, so they leak future refills), and device UDIs (they encode manufacture and expiry dates). Race and ethnicity are not used (D15; a LightGBM check showed they cost no measurable score). Three observation codes, DALY, QALY, and QOLS, are removed (D36): they are simulator scores recorded once a year from 2011, their counts predict an anchor before 2015 with AUROC 0.999, and they tell the model how the anchor was constructed rather than anything clinical. Removing them cost about 0.014 CV AUROC and probably costs test score too, since the test files contain them.

Genomics and the brain-MRI DICOM volumes are included at my request after an earlier decision to leave them out. Coverage is thin: genotype files exist for 396 training and 2 test patients, and a pre-anchor scan for 200 training and 28 test patients. The scan enters as one token only when its study date is before the anchor day. Measured in CV they add nothing (+0.0005 AUROC; on the 541 training patients whose inputs actually change, -0.003 [-0.020, +0.015]). The volumes are near-identical simulated brains (block-mean correlation 0.995 to 0.999 between patients).

Continuous values. Each numeric observation is binned into quantiles fitted per code and unit on the fold's fit patients only (up to 10 bins, merged where boundaries coincide), and the bin index is fused into the token. Units stay separate; no unit conversions or clinical thresholds come from outside. A value token needs at least five fit patients, otherwise it maps to `obs:code:other_value`. A LightGBM check compared continuous latest values against their bins and found no material loss (0.243 against 0.244 mAP), so the binned representation stayed.

Time between events. Every token carries the patient's age at that event, to the day, as `age/100` plus sine and cosine terms over eight periods from 0.5 to 200 years, mapped by a learned linear layer and added to the token embedding. The model can therefore read the gap between any two tokens. Rotary position encoding gives token order. The prediction head also receives the exact time from the last event to the anchor, age at anchor, sex, and a missing-history flag. Explicit gap tokens were planned as an experiment and not reached within the budget.

Context and summary. Sequences are capped at 1,024 tokens. The most recent complete event days are kept; conditions first diagnosed before the kept window become `hx:` summary tokens (up to 128) at the front, so an old diagnosis is never lost. Under the final tokenizer about 4.6% of training patients are truncated, but they are the densest histories: 15 to 18% of all event tokens fall outside the window, mostly medication refills and encounters. Exact refills (a stop and a restart of the same drug on the same day; 160,150 of them in fold 0) are merged first. Contexts of 512 and 2,048 were screened on fold 0 and did not help.

Leakage controls. Train and val timelines are cut with the same rules the released test files follow: keep rows dated before the anchor day, blank any `STOP` on or after it. The rule was chosen among four candidates by matching the distribution of the gap from last event to anchor in the test files (D1). The pipeline stops if any table holds a date on or after an anchor. A test adds a future record and checks that every prediction is unchanged. Labels come from the full record for train and val

patients; the prediction interface never sees them. The vocabulary, bins, folds, and self-pretraining windows are fitted inside each fold on fit patients only.

2. Model and training

Architecture. Decoder-only transformer with pre-LayerNorm blocks, causal multi-head attention, RoPE, GELU MLP (4x width), token embedding plus the age projection: 2 layers, width 128, 4 heads, vocabulary about 1,930 tokens, 738k parameters in total. The pretraining head predicts the next token with weights tied to the embedding; input-only symbols are masked out of the output distribution. The prediction head concatenates the last clinical-event state with the mean state over all clinical events (the `last_mean` readout adopted in round 2) and the four head features, applies dropout, and maps to 40 logits with independent sigmoids. The MRI encoder is a four-layer 3-D CNN (81k parameters) from random weights on a 64-cubed volume.

Objectives. Self-pretraining: next-token cross-entropy over clinical event positions, on each fit patient's history strictly before the real anchor, one random full window per patient per epoch; target-diagnosis tokens weigh 5 in the loss (round 2). Fine-tuning: unweighted binary cross-entropy over the unmasked patient-target cells (a target the patient already has is masked out of the loss and set to probability 0 at prediction), normalised by the cell count across gradient-accumulation micro-batches; one extra supervised example per fit patient per epoch from an earlier anchor five years back (round 2).

Hyperparameters (`configs/experiments/v2_c1.toml`): AdamW. Pretraining: lr 0.001, betas (0.9, 0.95), weight decay 0.1, 10% warm-up then cosine to 0.1 of peak, batch 32 patients, 30 epochs, patience 3 on fixed held-out windows; every run reached epoch 30 with held-out loss still falling slowly (fold 0: 1.449). Fine-tuning: head lr 0.001, body lr 0.0001, weight decay 0.01, gradient clip 1.0, batch 32, dropout 0.3 in the body and head, up to 30 epochs, patience 5 on held-out masked macro AUROC; best epochs ranged from 3 to 13. One fold (pretraining plus fine-tuning) takes about 20 minutes on a 16 GB M-series Mac; the 15 submitted members took about 4.5 hours in all. Everything is float32.

Three training methods were compared on the same folds before any tuning: S (random initialisation, supervised only), H (pretrained body held fixed, head only), and F (pretrained body, full fine-tuning). F won on every fold: mean fold AUROC 0.760 against 0.744 (S) and 0.742 (H); pooled paired difference F minus S +0.020 [+0.013, +0.028]. Self-pretraining helps; a fixed body does not carry enough.

Tuning, pre-registered. Two rounds, each written to a budget file and committed before any of its scores existed, with a fixed adoption rule: mean AUROC gain at least 0.005, higher on at least 4 of 5 folds, mAP loss at most 0.005. Round 1 (11 trials) adopted dropout 0.3; lower body learning rate, weight decay 0.1, an age-sex head, 60 pretraining epochs, and contexts 512 and 2,048 failed. Round 2 (12 trials, after removing the clock codes) adopted the combination of a pooled readout, extra anchors, and target-weighted pretraining; a count branch, per-target loss weights, and dropout 0.5 failed. Full-timeline pretraining (reading fit patients' post-anchor events) passed the rule and was **not** adopted, because the task says to truncate at the anchor. Three seeds of the winner were trained and averaged. The full register with every score is in `docs/decisions.md` (D35, D36) and `reports/transformer/`.

Curves. Training loss, learning rate, and gradient norm are logged per step, and held-out fold metrics per epoch, for all 199 runs in the wandb project above. Figure 1 shows the requested validation curves: every saved epoch checkpoint of the 15 members scored on the official val split after training,

inside the frozen bundle. The epochs selected on the held-out training folds (dots) sit near but mostly before the val optimum; val would have picked a later epoch in 13 of 15 members and gained 0.008 AUROC on average. That choice was not made.

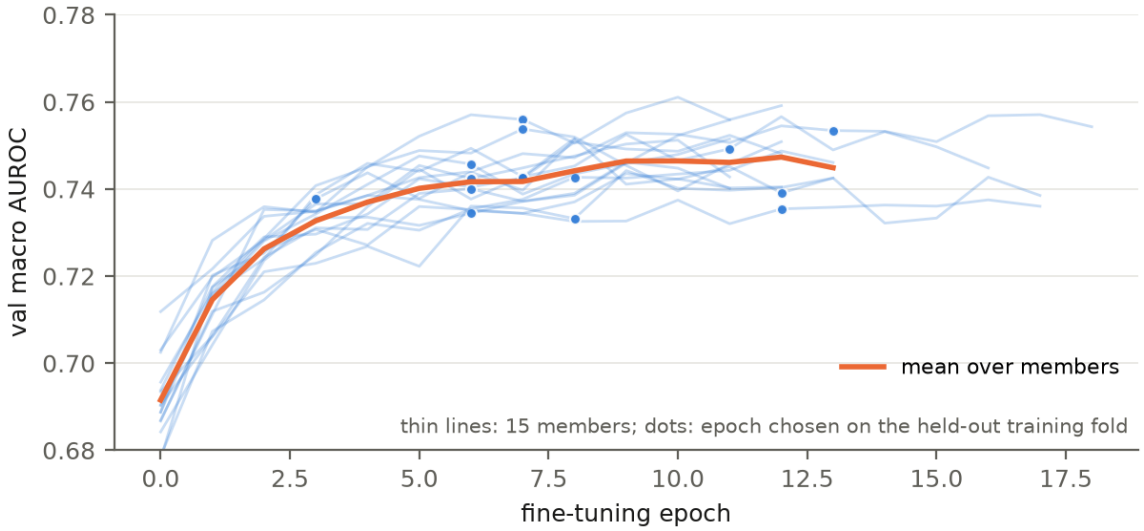


Figure 1: Official val macro AUROC of every saved fine-tuning checkpoint. Thin lines: the 15 ensemble members; dots: the epoch each member's held-out training fold selected; the orange line is the mean over members at that epoch.

Evaluation protocol. Five folds of the training split (patients never shared across folds), one out-of-fold prediction per patient from the model that did not fit on them, macro AUROC and mAP over 40 targets with the common prior mask, plus Brier score, with 95% patient-level BCa bootstrap intervals (1,000 resamples) and paired bootstraps for differences. Official val was scored once, at the end, by `make val-score`: it recorded 294 artifact, data, code, and lock-file hashes at freeze, refused to run on a changed tree or artifact, wrote a start record before any metric and a completion record after, and refuses to run again without a documented reason. Comparison models: an age-sex incidence table (the epidemiological floor, with and without the "already has it" flag), L2 logistic regression on log code counts, and LightGBM on counts and on counts plus latest values, each refitted on training patients with settings chosen by the same folds.

3. AI workflow

Claude Code wrote most of the code, the tests, and the planning documents; I set the objective and constraints, approved or rejected each decision that touched the data or the protocol, and read every result before the next step. The division is recorded in `docs/ai_workflow.md` and the reasoning for each decision in `docs/decisions.md`.

What kept the AI's work honest: the plan was written before the code and each revision says what was known when it was made; 175 unit tests include mutation-style checks (changing < to <= in the anchor rule fails a test; a future record must leave predictions unchanged; a resumed run must match an uninterrupted one); the adoption rule was fixed before the first transformer result; and every experiment block was committed before its scores.

Problems the review caught: fold-4 pretraining filled the disk with swap four times because the MPS

allocator fragmented across variable-length batches (fixed by releasing cached memory each step, verified to change no computation, and 3 to 6 times faster); LightGBM and PyTorch each loading an OpenMP runtime and crashing in one process (LightGBM now runs in its own process); the first bootstrap intervals using the percentile method, which for rare targets barely contained the point estimate (BCa replaced them); and a preflight audit flagging its own report files as a dirty tree. [Derek: add anything else here, including any use of Codex CLI.]

4. Interpretation

Where the model does well. Seven targets are above 0.95 AUROC for every model: prostate neoplasm, carcinoma in situ and metastasis of the prostate, normal pregnancy, pneumonia, obesity, and the pregnancy-complication code. Each follows a visible Synthea pathway (PSA tests and rectal exams, oral contraceptives, dementia care plans, BMI observations). A leakage review of these scores found pre-anchor causes, not leaks. The transformer's clearest win is the paired brain targets, silent micro-haemorrhage and abnormal gait (0.905 against 0.832 for LightGBM; the two labels are identical for every patient because one module emits both), and stroke (0.801 against 0.783). Figure 2 shows every target.

Where it does poorly, and why. Nine targets are below 0.6 for every model: whiplash, lacerations, wrist and ankle injuries, clavicle fracture, sinusitis, and strep throat. Synthea generates these acute events with little dependence on the prior record, so no history model can rank them; the age-sex table is at chance too. The transformer's largest losses to LightGBM are the count-driven conditions: osteoporosis (0.804 against 0.895), recurrent rectal polyp (0.860 against 0.946), myocardial infarction (0.827 against 0.904), the three lung-cancer codes (0.76 to 0.79 against 0.83 to 0.85), and pathological fracture (0.847 against 0.909). These were also the largest CV losses. In Synthea their risk grows with how many of a specific event has happened (screening visits, bone-density tests, nodule follow-ups), which a count feature states directly; the transformer has to infer it from a window that drops 15 to 18% of event tokens for exactly the dense histories where counts are high. The count branch tested in round 2 did not close the gap, and a longer context did not help on fold 0.

Anchor era. Anchors are five years before the last encounter, so an anchor before 2015 almost always means the patient died by 2020 (1,155 of 1,219 training patients), and those patients have more new diagnoses (2.37 against 1.48). Scores split by era are similar for both models (transformer 0.722 before 2015, 0.730 from 2015), and an era signal remains in the clinical codes and age even without the clock codes (AUROC 0.865). It cannot be removed without removing history, so it is reported.

Calibration. Prior-condition cells are exact zeros. Elsewhere the transformer's mean predicted probability is slightly below prevalence on most targets and its Brier score is 0.0011 worse than LightGBM's; the ensemble improved Brier over a single member by 0.0007.

Limits. The data are synthetic and the anchor is retrospective, so nothing here is evidence about clinical use. Val is small: the macro AUROC interval is 0.04 wide and nine targets have at most eight positives. The CV scores of the chosen configuration are the best of 23 trials on the same folds, so they are optimistic; the val score is the first unbiased one, and the hidden test score is the second. Two seeds of CV variation (about 0.002 AUROC in the mean, up to 0.014 per fold) are comparable to several of the differences in the tuning table.

5. Results summary and reproduction

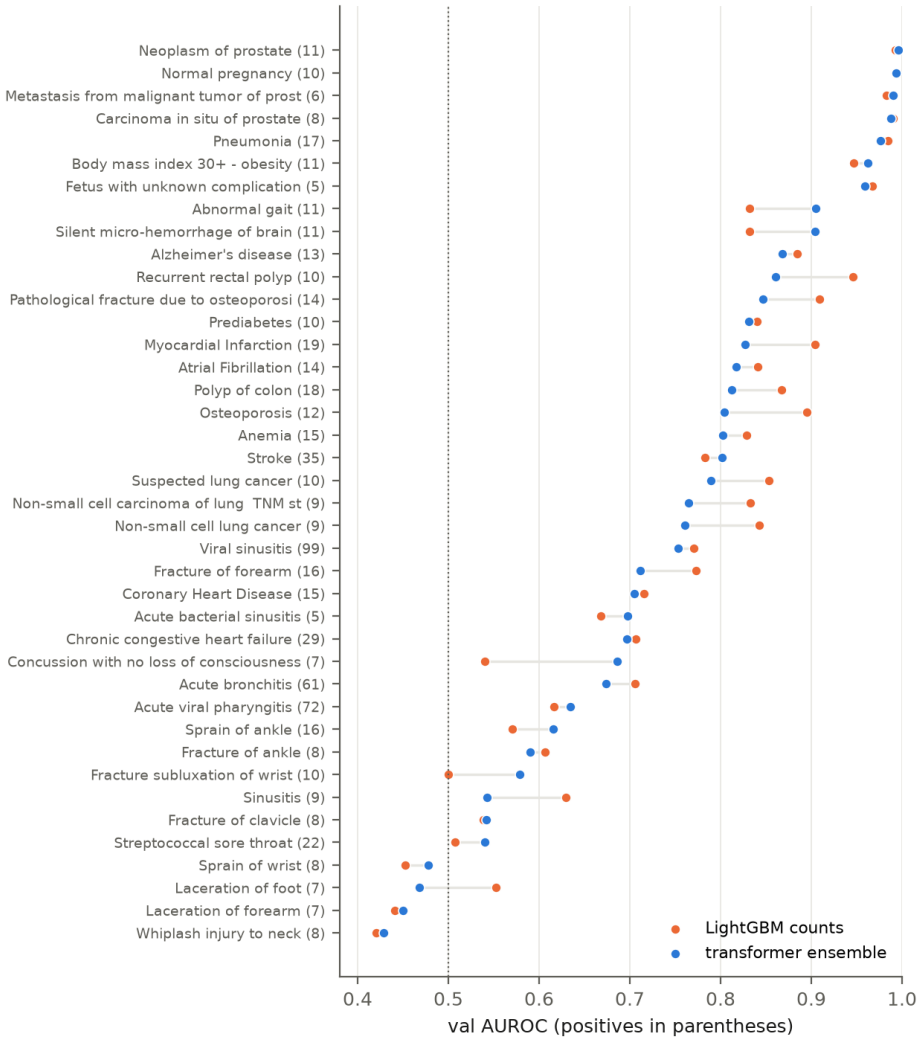


Figure 2: Official val AUROC per target, transformer ensemble against LightGBM on code counts, sorted by the transformer. Numbers in parentheses are val positives; most targets have 5 to 20, so single- target differences are noisy.

Split	Model	Macro AUROC	mAP
train CV, pooled out of fold	transformer ensemble	0.761	0.181
train CV, pooled out of fold	LightGBM counts	0.764	0.188
val (365)	**transformer ensemble**	**0.751** [0.730, 0.773]	**0.209** [0.192, 0.211]
val	logistic regression, codes	0.764	0.222
val	LightGBM counts	0.762	0.239
val	LightGBM counts + values	0.758	0.257
val	age-sex incidence, at risk	0.727	0.141
test (358)	transformer ensemble	organisers' score	organisers' score

Paired on val, transformer minus LightGBM counts: AUROC -0.010 [-0.031, +0.010], mAP -0.030 [-0.060, -0.011]. `predictions.csv` holds 358 rows by 40 code columns from the same 15 members.

Reproduction: `make setup`, link the dataset to `data/raw`, then `make dataset`, `make modalities`, `make folds`, `make tokenize`, `make pretrain` and `make finetune` per fold and seed, `make freeze`, `make val-score`, `make predict`. The Hugging Face package carries the 15 checkpoints and 5 tokenizers in the layout the prediction command reads; `python -m ehrforecast.hf_package verify` reloads it on CPU and reproduced `predictions.csv` exactly, both in the development environment and in a fresh virtual environment built from `requirements-lock.txt`.